Perfusion Index and Pulse Oximetry Screening for Congenital Heart Defects

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Objective To evaluate the efficacy of combined pulse oximetry (POX) and perfusion index (PI) neonatal screening for severe congenital heart defects (sCHD) and assess different impacts of screening in tertiary and nontertiary hospitals.

Study design A multicenter, prospective study in 10 tertiary and 6 nontertiary maternity hospitals. A total of 42 169 asymptomatic newborns from among 50 244 neonates were screened; exclusion criteria were antenatal sCHD diagnosis, postnatal clinically suspected sCHD, and neonatal intensive care unit admission. Eligible infants underwent pre- and postductal POX and PI screening after routine discharge examination. Targeted sCHD were anatomically defined. Positivity was defined as postductal oxygen saturation (SpO_2) ≤95%, prepostductal SpO_2 gradient >3%, or PI <0.90. Confirmed positive cases underwent echocardiography for definitive diagnosis. Missed cases were identified by consulting clinical registries at 6 regional pediatric heart centers. Main outcomes were incidence of unexpected sCHD; proportion of undetected sCHD after discharge in tertiary and nontertiary hospitals; and specificity, sensitivity, positive predictive value, and negative predictive value of combined screening.

Results One hundred forty-two sCHD were detected prenatally. Prevalence of unexpected sCHD was 1 in 1115 live births, similar in tertiary and nontertiary hospitals. Screening identified 3 sCHD (low SpO_2, 2; coarctation for low PI, 1). Four cases were missed. In tertiary hospitals, 95% of unsuspected sCHDs were identified clinically, whereas only 28% in nontertiary units; in nontertiary units PI-POX screening increased the detection rate to 71%.

Conclusions PI-POX predischarge screening provided benefits in nontertiary units, where clinical recognition rate was low. PI can help identify coarctation cases missed by POX but requires further evaluation in populations with higher rates of missed cases. *(J Pediatr 2017;183:74-9).*

Congenital heart defects (CHDs) are the most common neonatal malformations, with a prevalence of 0.8%-1%, and are a leading cause of infant death in developed countries. A significant proportion of CHD (1.5/1000 live births) are potentially fatal during the neonatal period.[1] Early recognition of severe CHD (sCHD) is highly important, as surgical correction or palliation is possible for a certain proportion of cases and may improve survival and overall quality of life.

Although advances in prenatal medicine have increased the number of CHD detected through antenatal ultrasound screening, several cases still remain unidentified at birth.[2-4] In addition, as critical CHD often causes subtle symptoms in the early days of life, some cases may be missed during routine neonatal evaluations. Presymptomatic CHD diagnosis is thought to allow better perioperative patient management, reduce hospital stays, and improve survival and prevent neurologic sequelae secondary to hypoxia and acidosis.[5,6]

CHDs	Congenital heart defects	POX	Pulse oximetry
HLHS	Hypoplastic left heart syndrome	sCHD	Severe CHD
HoL	Hours of life	SpO_2	Oxygen saturation
PI	Perfusion index		

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Associazione Genitori Neonati ad Alto Rischio (AGENAR) financially supported the realization of the electronic database platform. AGENAR had no role in study design, in the collection, analysis, and interpretation of data, in the writing of the report, and in the decision to submit the paper for publication. The authors declare no conflicts of interest.

Portions of the study were presented at the Congress of the European Academy of Paediatric Societies, October 17-21, 2014, Barcelona, Spain.

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http://dx.doi.org10.1016/j.jpeds.2016.12.076

Pulse oximetry (POX) has gained increasing interest in the last decade as an early CHD screening tool in neonates and in 2011 POX screening for critical CHD was included in the recommended uniform screening panel in the US.[7] Other countries are evaluating the potential benefits of implementing this type of screening.[8-12]

The reported sensitivity and specificity of POX vary according to the evaluation timing, adopted cut-off value, use of a single or paired measurement site (postductal only or pre- and postductal), and evaluated target lesion type.[13-18] However, all previous studies described a low sensitivity for left ventricle outflow obstructive defects, including aortic coarctation, interrupted aortic arch, critical aortic stenosis, and hypoplastic left heart syndrome (HLHS), as the main limitation of POX.[14,16-21]

Left-sided heart obstructions are the lesion type most frequently missed by POX and during routine physical examinations.[22,23] Accordingly, POX screening might be ineffective for the very lesions that most require additional diagnostic efforts to improve their detection rate.[24,25]

In 2007, Granelli and Ostman-Smith[26] proposed the peripheral perfusion index (PI) as a possible screening tool for congenital left heart obstruction. PI is a POX-derived, oxygen saturation-independent measure expressed as the ratio of the pulsatile component of the infrared light signal that reaches the sensor (reflected by the arterial blood) to the nonpulsatile component (reflected by the venous blood and tissues) and, thus, represents the relative amount of arterial perfusion in the examined area.[27] In neonates, PI correlates with other measures of peripheral and systemic perfusion obtained via functional echocardiography or near-infrared spectroscopy.[28,29] Granelli and Ostman-Smith[26] observed PI values below the fifth percentile in 6 of 9 infants with a congenital left heart obstruction. However, although promising, the effectiveness of a PI-based screening protocol has not yet been studied prospectively.

Aims of the study were to assess the prevalence of unexpected (not suspected prenatally) sCHD in tertiary and nontertiary hospitals and to evaluate the efficacy of neonatal screening for sCHD using combined POX and PI at the time of patient discharge.

Methods

This study was approved by the ethics committee of the coordinating center. Written consent was obtained from parents at centers where POX screening was not routinely performed.

All neonates who were admitted to well-baby nurseries from June 2011 to November 2013, were asymptomatic at predischarge examination, and had not received a previous echocardiographic assessment were eligible. Patients were enrolled consecutively at each center, with variable starting and ending dates. Infants admitted to the neonatal intensive care unit, suspected prenatally of CHD, or presenting conditions requiring cardiac assessment (genetic syndromes, major extracardiac anomalies, family history of CHD, mother with diabetes) were excluded. Screening occurred at 48-72 hours of life (HoL) at the time of metabolic screening. None of the participating hospitals adopted a policy of early discharge (<48 HoL). Oxygen saturation (SpO_2) and PI were measured in both preductal (right hand) and postductal sites (foot).

To avoid motion artifact effects, PI readings were taken after a regular sinusoidal track was displayed for 10 seconds. Masimo neonatal disposable sensors (Masimo Corporation, Irvine, California) were used along with either the recommended Masimo Rad 7pulse oximeter or Bitmos Sat 805 (Bitmos GmbH, Dusseldorf, Germany) oximeter, which uses Masimo SET technology and displays PI similarly. Positive screening tests met one of the following conditions: postductal SpO_2 ≤95%; >3% difference between pre- and postductal SpO_2; or single-site PI <0.90. Positive screening tests were repeated after 30 minutes for confirmation. Patients who failed the second test were referred for same-day echocardiography.

We selected a higher PI threshold than that originally proposed by Granelli and Ostman-Smith[26] (0.70) based on our preliminary observation that the PI was slightly higher after 48 HoL than during the first 48 hours (data not shown), consistent with previously published data.[30] This threshold was intended to minimize false negative cases while maintaining an acceptable false positive rate. In addition, assuming that combining PI and POX could reduce the number of missed cases from 2.5 to 0.62 per 10 000 neonates (75% reduction), we calculated that 40 000 neonates should be screened to draw statistically significant conclusions (0.80 power, 0.05 alpha error).

Target lesions were anatomically defined severe congenital heart malformations (sCHD; **Table I**). Simple cardiac lesions (eg, isolated ventricular septal defect, atrial septal defect, patent foramen ovale, patent ductus arteriosus, bicuspid aortic valve, mild pulmonary stenosis, and mild aortic stenosis) were excluded. With the aim of exploring the full potential of the new screening protocol the list of potential targets was somewhat broader than those used in other studies.

Table I. Cardiac anomalies defined as target lesions of PI-POX screening

IAA, interrupted aortic arch*
AS, aortic stenosis (moderate to severe)
AVC, atrioventricular canal
CoA, aortic coarctation*
DORV, double outlet right ventricle*
Ebstein anomaly*
HLHS, hypoplastic left heart syndrome*,†
PA/VSD, pulmonary atresia with ventricular septal defect*,†
PA/IVS, pulmonary atresia with intact ventricular septum*,†
PS, pulmonary stenosis (moderate to severe)
PTA, persistent truncus arteriosus*,†
SV, single ventricle*
TA, tricuspid atresia*,†
TAPVR, total anomalous pulmonary venous return*,†
d-TGA*,†
ToF, tetralogy of Fallot*,†
Other complex or rare CHDs with cyanosis and/or low systemic perfusion*

d-TGA, d-transposition of the great arteries.
*Defects indicated as core targets by the Newborn Screening for Critical Congenital Heart Disease Expert Panel (CDC-AAP).[31]
†Defects indicated as pulse oximetry primary target by the US Secretary's Advisory Committee on Heritable Disorders in Newborns.

POX and PI results and echocardiography diagnoses of positive cases were reported in an online electronic database. All sCHD cases and details regarding the diagnostic modality (prenatal, clinical postnatal, screening, or clinical postdischarge) were also reported.

Possible missed cases were identified after a 3-month minimum follow-up after study completion by consulting the clinical registries of 6 pediatric cardiology/heart surgery centers near the participating hospitals. To identify possible early deaths from undiagnosed sCHD, the national sudden infant death syndrome and sudden unexpected infant death data bank at the University of Milan was also searched.

Screening accuracy was assessed as the sensitivity and specificity (with corresponding 95% CIs) in the whole study population and separately in the 2 groups of hospitals (tertiary and nontertiary). Fisher exact test was used to compare the 2 groups of hospitals. Statistical significance was set at P value of < .05.

Results

During the study period, 50 244 infants were born in the participating maternity units. Of these, 8075 were excluded from screening or analysis because of neonatal intensive care unit admission, prescreening cardiac assessment, parental refusal, temporary unavailability of the recommended device, staff shortage, or incomplete data reporting. Among nonscreened newborns, 180 cases of sCHD were diagnosed before discharge: 142 cases were diagnosed via antenatal ultrasound screening, 33 from clinical signs of CHD or abnormal routine physical examinations, and 5 following investigation for genetic syndromes or extracardiac anomalies.

Finally, 42 169 asymptomatic newborns were screened, of whom 618 were initially positive and 194 remained positive during the repeated test and were referred for echocardiography. Echocardiography subsequently excluded sCHD in 191 cases (false positive), of which 23 presented with a minor cardiac problem (atrial septal defect, ventricular septal defect, patent ductus arteriosus, or persistent fetal circulation); sCHD was identified in 3 infants, 2 of whom had a SpO_2 below the threshold (1, tetralogy of Fallot, 1, cor triatriatum dexter), and 1 infant had aortic coarctation and a low postductal PI (0.3) with normal SpO_2.

After the follow-up, 4 patients who had passed the screening and were discharged were found to have sCHD (false negative): 2 cases of aortic coarctation, 1 case of truncus arteriosus with interrupted aortic arch, and 1 case of left pulmonary artery agenesis with aortopulmonary collaterals (see **Table II** for the complete list of identified cardiac defects and the corresponding diagnostic modality).

Including these patients, the prevalence of sCHD in newborns without a prenatal diagnosis of cardiac disease was 1 in 1115 live births.

In addition, 169 protocol violations were registered; in most cases, the test was not repeated after an initial positive result, whereas in 9 cases echocardiography was not performed after a repeated positive screening. None of these patients presented with sCHD during the follow-up.

Table II. Identified cardiac anomalies and diagnostic modalities

Cardiac lesions	Prenatal	Clinical postnatal	Screening	Postdischarge	Total
IAA	2	1	—	1	4
AS	3	—	—	—	3
AVC	11	5	—	—	16
CoA	21	4	1	2	28
DORV	12	1	—	—	13
Ebstein anomaly	8	—	—	—	8
HLHS	10	—	—	—	10
PA/VSD	7	—	—	—	7
PA/IVS	5	—	—	—	5
PS	4	9	—	—	13
PTA	3	1	—	—	4
SV	1	—	—	—	1
TA	2	—	—	—	2
TAPVR	—	5	—	—	5
TGA	28	6	—	—	34
ToF	25	5	1	—	31
Other	—	1	1	1	3

The screening results are summarized in **Figure 1** (available at www.jpeds.com). The combined screening sensitivity, specificity, positive predictive value, and negative predictive value were 42.8% (95% CI 6.1%-79.5%), 99.54% (99.48%-99.60%),1.54% (0.19%-3.28%), and 99.99% (99.99%-99.99%), respectively. The false positive rate was 0.45% (0.27% for PI), although 1 center had an exceptionally high false positive rate of 6%. After excluding this center from analysis, the false positive rate, specificity, and positive predictive value were 0.15%, 99.81%, and 3.94%, respectively.

During the study period, 23.6% of the neonates (n = 11 844) were born in nontertiary (first or second level) maternity units. Although most (139/142) neonates with prenatally detected sCHD were delivered at tertiary centers, the prevalence of unsuspected sCHD was similar in both groups of hospitals: 1 in 1010 live births in tertiary hospitals vs 1 in 1692 in nontertiary hospitals ($P = .2$).

sCHD was suspected clinically before screening in 36 of 38 cases (95%) at tertiary hospitals but in only 2 of 7 cases (28.5%) at nontertiary hospitals ($P < .001$). Although PI-POX screening failed to identify 2 clinically missed cases in tertiary hospitals, it detected 3 of 5 nearly undiagnosed cases at nontertiary hospitals (**Figure 2**). The PI-POX screening sensitivity in nontertiary hospitals was 60% (CI 23.1%-88.2%). False positive rate was higher in tertiary hospital (0.53% vs 0.18%), but after exclusion of the outlier center, both groups of hospitals had similar false positive rates (0.20% vs 0.18%, $P = .4$).

Of the 35 cases of critical left heart obstruction (aortic coarctation = 22, HLHS = 6, aortic arch interruption = 4, aortic stenosis = 3), 29 were suspected at the antenatal ultrasound screening. Two of the 6 cases unknown at birth were identified early based on clinical signs (1 syndromic case); the remaining 4 patients were screened. Although none had SpO_2 below the threshold, 1 case of aortic coarctation was identified because of a persistently low postductal PI (0.3). Two of the 3 missed cases were readmitted soon after discharge with symptoms of shock or cardiac failure (aortic coarctation,

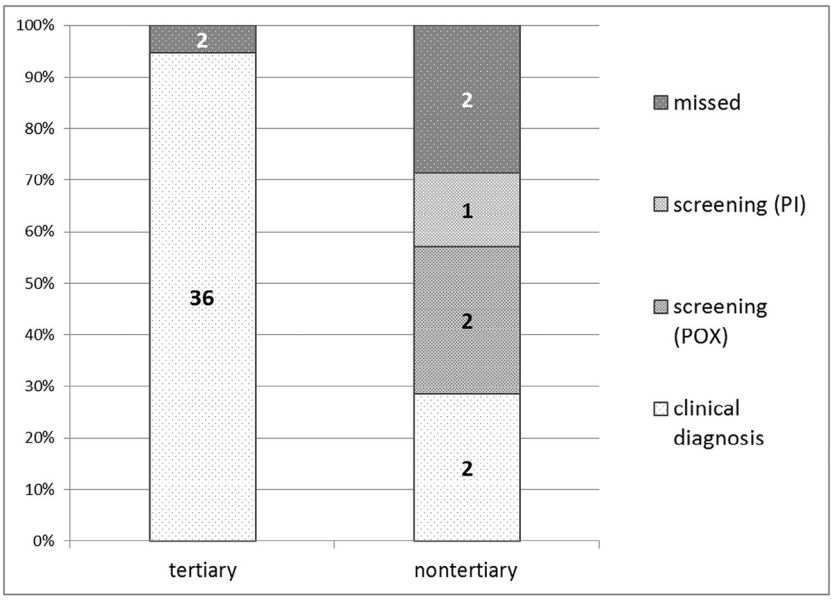

Figure 2. Diagnostic modalities for prenatally undiagnosed sCHD in tertiary and nontertiary maternity units.

interrupted aortic arch/persistent truncus arteriosus). The family doctor identified 1 case of aortic coarctation because of weak femoral pulses at 2 weeks of age.

Discussion

One of the most notable findings to emerge from this study is the elevated number of sCHD detected from prenatal ultrasound scans or clinical symptoms before predischarge screening (76% cases before birth and 96% before discharge). Although the high prenatal detection may be overestimated, as some cases might have been identified at peripheral nonparticipating centers and referred to a participating tertiary center for a protected birth, nonetheless, the prevalence of unknown sCHD in the neonatal population is not affected by in utero transfer and can be taken as a reliable measure of local prenatal screening efficiency, assuming negligible geographic variability in the sCHD incidence. Accordingly, we found that 9 of 10 000 neonates presented with sCHD that was unknown at birth. This finding agrees with series published in Germany, Norway, and the US and is better than recently reported rates in Sweden, United Kingdom, and China.[13-17,32]

Good prenatal screening performance was confirmed by the high detection rate for specific lesion types, including all univentricular heart (HLHS, tricuspid atresia, single ventricle), Ebstein anomaly, and pulmonary atresia cases. High prenatal detection rates were also recorded for d-transposition of the great arteries (82%) and aortic coarctation (75%). These percentages are superior to those reported by Wren in 2008 and by most of the later studies.[2,33-35] Despite these encouraging results, we found that approximately 1 in 1000 infants was affected by nonprenatally detected sCHD. In most such cases (38/45, 84%), clinical symptoms or routine physical examination led to a diagnosis before the planned predischarge screening. In previous large multicenter studies of POX screening, the proportion of patients identified before the screening varied from 38% to 62.5%; only Zhao et al[32] reported a clinical detection rate (81%) similar to ours. Conversely, POX screening was ineffective in some recent single-center studies because all affected cases were suspected from clinical signs.[36-38]

Several factors may have contributed to the high clinical detection rate in our study and, therefore, limited the diagnostic gap that POX screening was expected to fill. In our study, POX screening was scheduled to occur after the routine discharge physical examination. In most previous studies, neonates were screened during the first or second day of life. Although early screening might provide some advantages (anticipated diagnosis within a few hours, early recognition of noncardiac disease), the screening test yields might be overestimated because an equal frequency of "true positive" cases would have been identified clinically at a slightly later time.[39] Moreover, we screened newborns at 48 hours because in contrast to Northern Europe and the US, Italy does not have a common policy of discharge before 48 HoL, at which time cardiac disease is more likely to have become apparent.

An important finding of the present study was the considerably higher clinical sCHD detection rate in tertiary hospitals relative to nontertiary hospitals. Moreover, the 2 cases missed during routine physical examination at tertiary centers also escaped PI-POX screening detection. Conversely, at nontertiary hospitals PI-POX screening identified 3 of 5 clinically missed cases. Other studies showed that the POX test was more relevant in settings with low prenatal detection rates, and 1 study showed that infants born in a level I or level II nursery were more likely to have late-detected critical congenital heart disease compared with infants born in a level III nursery.[15,38,40]

The present study confirms the variable impact of POX screening in centers with different experiences and volumes of activity, being particularly beneficial in nontertiary maternity hospitals. The 6 nontertiary participating centers reported lower annual birth numbers as well as lower absolute and relative numbers of CHD cases relative to tertiary centers (1/1200 vs 1/200 neonates). Moreover, in nontertiary centers, medical staff is rarely dedicated to neonates and is instead usually shared with the general pediatrics ward. Our data support the hypothesis that medical staff with little cardiac case experience may exhibit poorer expertise and alertness to heart disease, leading to an increased risk of missing a prenatally undetected congenital malformation; in this setting POX screening provides significant assistance. In Italy, approximately 200 000 neonates are born annually in units experiencing fewer than 1000 annual deliveries[41]; we can roughly estimate that annual nationwide screening of these small units might identify nearly 50 missed cases.

Our novel screening protocol featured the addition of PI to POX to reduce the rate of missed left heart obstruction cases. Although the overall number of clinically missed cases was smaller than anticipated, our data confirm that left heart obstruction (particularly aortic coarctation) is the most frequently missed lesion type during both prenatal screening and postnatal clinical assessments. POX failed to detect all screened coarctations, one of which was detected by PI. Because of the small number of screened cases, we cannot derive precise information about the sensitivity of PI. Nevertheless, we demonstrated that PI confers additional sensitivity upon POX regarding the most difficult-to-detect lesions while requiring no additional staff time and an acceptable increase in unnecessary echocardiograms. As the only identified case of coarctation had a PI far below the threshold, whereas the 2 missed coarctations had a completely normal PI, we can speculate that a lower threshold may yield a similar sensitivity and significantly lower false positive rate. Other proposed screening methods (eg, blood pressure measurement) have been recently rejected because of excessively high false positive rates.[42]

Unfortunately, the 2 missed coarctations prevent us from regarding PI as the ultimate method for timely congenital aortic obstruction detection, and we must accept that no current screening method (prenatal or postnatal) can guarantee that all infants in large populations are diagnosed before discharge. Accordingly, practitioners and parents should be well informed about this risk.

There were limitations to our study. Our protocol did not account for the training of specific healthcare staff or the professionals deputized to perform screening tests. Evidence suggests that these factors affect the measurement reliability, at least when using older oximeters.[43]

The high number of protocol violations and high false positive rate observed at 1 center might have been related to the absence of a preliminary training phase. Nonetheless, we believe that using new generation motion tolerant pulse oximeters, providing clear written indications for test performance, and repeating positive tests should have sufficiently limited errors due to human factors. The false positive rate was somewhat higher than expected. This may be partly explained by the high cut-off chosen for SpO_2 (<96%) and PI (<0,9) and by the use of a single confirmatory test. All centers were located at an elevation <600 m above sea level, therefore, it is unlikely that altitude played a role in the false positive rate.

We cannot exclude the possibility that isolated missed CHD cases were recognized and treated outside the geographic study area and, therefore, escaped our follow-up. In addition, the National Register of sudden infant death syndrome and sudden unexpected infant death is currently deficient, and we might, therefore, have missed single cases of death at home.

Our study demonstrated that the PI is able to detect at least some cases of coarctation missed by POX but requires further evaluation in populations with higher rates of missed cases. Neonatal screening for CHD provides a more relevant contribution in nontertiary settings where the risk of missed diagnosis is higher. ■

We thank the participating families and nurses who contributed to the study.

Submitted for publication Sep 10, 2016; last revision received Nov 9, 2016; accepted Dec 30, 2016

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Appendix

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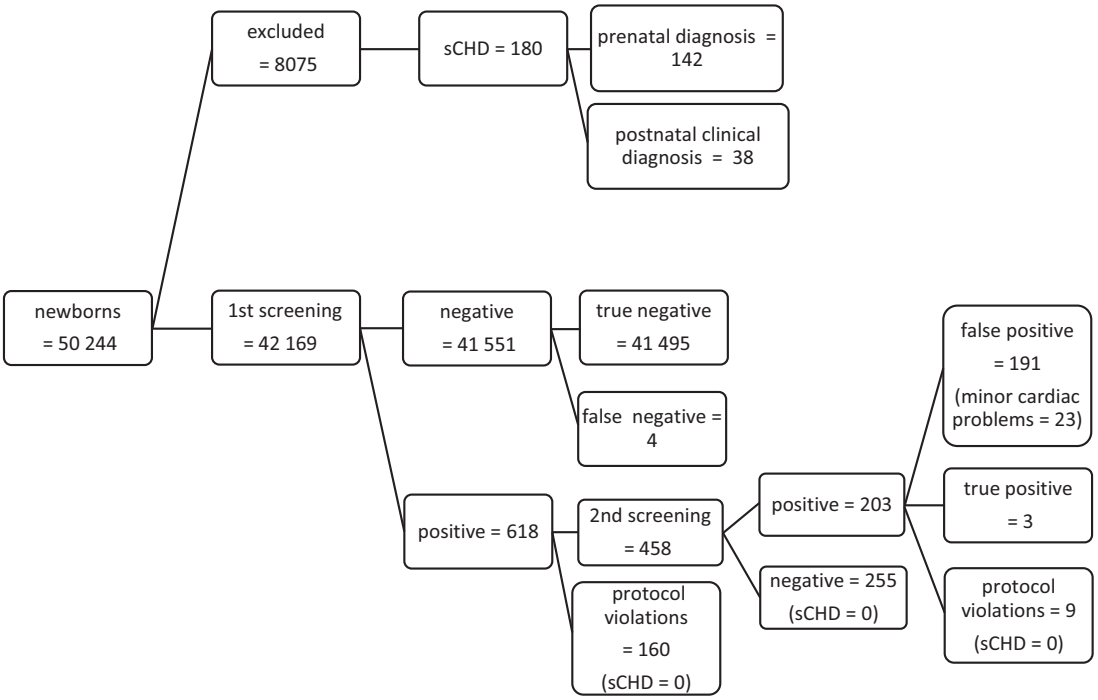

Figure 1. Study flow diagram and results.